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Article

Cite This: Ind. Eng. Chem. Res. 2019, 58, 14991−15002

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Application of the Preferential Solvation Viscosity Model to Binary Liquid Mixtures: Aqueous, Nonaqueous, Ionic Liquid, and Deep Eutectic Solvent Systems

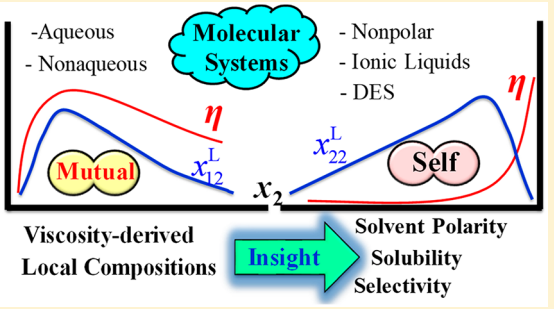
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Supporting Information

ABSTRACT: The preferential solvation (PS) viscosity model was used to correlate binary liquid viscosities of aqueous, nonaqueous, nonpolar−polar, molecular solvent−ionic liquid (IL), and molecular solvent−deep eutectic solvent mixtures for the purpose of determining local composition in the chemical systems and analysis of chemical phenomena. Lignin solubility in aqueous systems was directly proportional to the population of (1−2) complex molecules. Cellulose solubility in molecular solvent−IL mixtures was directly proportional to the population of (2−2) associated molecules. Local compositions determined from the PS viscosity model were able to be used to estimate mixture solvent polarity and the variation of carbazole selectivity in crude anthracene separations. The PS viscosity model applied to chromatography allowed estimation of appropriate bulk composition regions for separations. Local compositions determined from the PS viscosity model can be used to analyze a wide variety of chemical phenomena in liquid solvent mixtures.



1. INTRODUCTION

Binary liquid mixtures have many advantages over pure solvents in chemical reaction and separation processes.[1−3] For example, a properly chosen binary mixture of safe solvents can be used to replace hazardous solvents through methodologies based on physicochemical properties (e.g., Kamlet−Taft solvatochromic parameters[4−6] and Hansen solubility parameters[7]). The bulk composition of a mixture of two solvents, "1" and "2" that have specific (1−2) molecular interactions is a convenient means to control selectivity or solvation effects in a reaction or separation.[8−10] However, the properties of binary liquid mixtures often show deviations from the mole fraction average of pure component liquid properties that are attributed to molecular interactions. Methods to quantify the specific molecular interactions in binary liquid mixtures are needed to allow one to take advantage of property changes that occur with bulk composition.

Viscosity is a key transport property that is needed to estimate fluid properties and mass transfer in chemical systems. Due to the complexity of molecular interactions, semitheoretical models such as Eyring theory[11] and free volume theory[12,13] are commonly applied to correlate liquid viscosities of binary mixtures. When low deviations in viscosity correlations are required for engineering design, modified Eyring viscosity models, McAllister,[14−16] Jouyban−Acree,[17,18] or Grunberg−Nissan[19,20] models are used in which the adjustable parameters may be qualitatively related to the molecular features of a system. In previous work,[21] solvent exchange theory was used to develop a viscosity model based on preferential solvation (PS). The PS viscosity model allows the determination of local composition from bulk viscosity measurements and can be used to study variations in the population of (1−2) complex molecules or (1−1) or (2−2) associated molecules.[21]

Local composition of binary mixtures is generally accessible using spectroscopic techniques or molecular simulations; however, these approaches require detailed analyses and the specific interactions determined may not be directly related to the macroscopic property or the bulk composition. On the other hand, features of the hydration shell in aqueous mixtures or solvation shell in nonaqueous mixtures along with local composition can be quantified with the PS viscosity model through viscosity measurements.[21]

In this work, the PS viscosity model was applied to five types of molecular systems made up of binary liquid mixtures of (i) aqueous solutions of hydrogen bond acceptor solvents, (ii)

Received: March 2, 2019
Revised: June 14, 2019
Accepted: July 18, 2019
Published: July 18, 2019

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DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991−15002

Table 1. Comparison of Four Viscosity Model Correlations for Five Types of Molecular Systems Having Molecular (i–j) Interaction Type and N Data Points with a Maximum Mixture Viscosity of η_{max}[a]

				preferential solvation		McAllister		Jouyban–Acree		Grunberg–Nissan	
molecular system	i–j	N	η_{max} (mPa·s)	ARD (%)	R^2	ARD (%)	R^2	ARD (%)	R^2	ARD (%)	R^2
aqueous of HBD (1)–HBA (2)	1–2	718	5.1	1.030	0.996	2.146	0.980	2.316	0.977	17.85	0.206
nonaqueous of HBD (1)–HBA (2)	1–2	803	21	0.542	0.997	0.258	0.998	0.273	0.999	1.681	0.969
nonpolar (1)–polar (2)	2–2	387	2.6	0.588	0.997	0.261	0.998	0.289	0.999	0.799	0.995
molecular solvent (1)–IL (2)	2–2	529	2.1×10^4	3.911	0.999	3.263	0.999	2.916	0.999	9.927	0.986
molecular solvent (1)–DES (2)	2–2	112	1.2×10^3	2.517	0.999	2.201	0.999	2.798	0.999	7.899	0.998
overall		2549		1.717		1.626		1.718		7.631	

[a]Statistics shown are the average relative deviation (ARD) defined by eq 15 and coefficient of determination (R^2).

nonaqueous solutions containing a hydrogen bond donor (HBD) and a hydrogen bond acceptor (HBA) solvent, (iii) nonpolar molecule–polar molecule solutions that are completely miscible, (iv) solutions of a molecular solvent with an ionic liquid (IL), and (v) solutions of a molecular solvent with a deep eutectic solvent (DES). In this work, the DES was considered as a pseudopure component, even though a DES is a mixture of an HBD solvent and an HBA solvent. The PS viscosity model was compared with the McAllister, Jouyban–Acree, and Grunberg–Nissan viscosity models to explore parameter relationships. Local compositions obtained from the PS viscosity model were used to analyze several applications and to identify relationships between the local composition and the chemical phenomenon.

The objectives of this work were to (i) apply the preferential solvation viscosity model to correlate liquid viscosities of five types of molecular systems, (ii) analyze PS viscosity model parameters and their trends in molecular systems, (iii) examine relationships between PS viscosity model parameters with other viscosity model parameters, and (iv) examine four chemical applications related to cellulose solubility, lignin solubility, fractionation of carbazole from crude anthracene, and chromatographic mixed-solvent properties and analyze relationships between viscosity-derived local compositions and chemical phenomena in those applications.

2. VISCOSITY MODELS

2.1. Preferential Solvation Viscosity Model. The preferential solvation viscosity model assumes the existence of mutual (ij) complex molecules or (ii or jj) associated molecules formed from solvent i (S_i) and solvent j (S_j) due to the presence of strong specific molecular interactions, such as those resulting from hydrogen bond donor–acceptor functional groups. Solvation equations for several possible interactions are

$$(S_i^L)_m + mS_j \overset{g_{j/i}}{\rightleftarrows} (S_j^L)_m + mS_i \tag{1}$$

$$(S_i^L)_m + \frac{m}{2}S_{ij} \overset{g_{ij/i}}{\rightleftarrows} (S_{ij}^L)_m + \frac{m}{2}S_i \tag{2}$$

$$(S_i^L)_m + \frac{m}{2}S_{jj} \overset{g_{jj/i}}{\rightleftarrows} (S_{jj}^L)_m + \frac{m}{2}S_i \tag{3}$$

The general form of the PS viscosity model is

$$\eta_{mix} = \sum_{i=1}^{N} x_i^L \eta_i^0 + \sum_{i=1}^{N}\sum_{j=i}^{N} x_{ij}^L \eta_{ij} \tag{4}$$

where, η_{mix} is the mixture viscosity, η_i^0 are the pure component (bulk) viscosities, N is the number of distinct molecular compounds, and the "L" superscript refers to the local composition of a solvent molecule. Subscripts denote a component i or j and a complex molecule subscript ij = 12 or of an associated molecule (ii = 11 or jj = 22). For binary liquid mixtures, eq 4 becomes

$$\eta_{mix} = x_i^L \eta_i^0 + x_j^L \eta_j^0 + x_{ij}^L \eta_{ij} + x_{ii}^L \eta_{ii} + x_{jj}^L \eta_{jj} \tag{5}$$

Molecular interactions of the ij type or of the jj type are the main contribution to specific interactions in the five molecular systems of binary liquid mixtures studied. For aqueous solutions of a dipolar protophilic solvent (2) and for nonaqueous solutions of an HBD (1) and HBA (2) solvent, eq 5 becomes

$$\eta_{mix} = x_1^L \eta_1^0 + x_2^L \eta_2^0 + x_{12}^L \eta_{12} \tag{6}$$

where x_{12}^L is the local composition of the (12) complex molecule and η_{12} is related to the viscosity of the (12) complex molecule.

For solutions of nonpolar (1)–polar (2) molecules, the (2–2) self-interaction dominates the specific interactions in the molecular system, so that for nonpolar molecule–polar molecule solutions, eq 5 becomes

$$\eta_{mix} = x_1^L \eta_1^0 + x_2^L \eta_2^0 + x_{22}^L \eta_{22} \tag{7}$$

Molecular solvent (1)–IL (2) and molecular solvent (1)–DES (2) systems have similar trends in viscosity as those for nonpolar (1)–polar (2) systems, because even though molecular solvent–IL systems have more than one type of specific interaction, (2–2) self-association of the ionic liquid is the chief specific interaction.[22,23] Similarly, for molecular solvent (1)–DES (2) systems, the (2–2) self-association of the DES is regarded to be the chief specific interaction in the molecular system, so that eq 7 is used in the analyses for both molecular solvent–IL and molecular solvent–DES systems. The main (i–j) interactions of five types of molecular systems are summarized in Table 1.

The PS viscosity model is

$$\eta_{mix} = \eta_1 + \frac{g_{2/1}(\eta_2 - \eta_1)(x_2)^m + g_{ij/1}(\eta_{ij} - \eta_1)[(1 - x_2)x_2]^{m/2}}{(1 - x_2)^m + g_{2/1}(x_2)^m + g_{ij/1}[(1 - x_2)x_2]^{m/2}} \tag{8}$$

where $g_{2/1}$, $g_{ij/1}$, and η_{ij} are preferential solvation parameters in which the average number of molecules (m) in a complex molecule (eqs 1, 2, and 8) is set equal to 2.5 according to results from previous work[21] and based on preliminary assessment on five types of molecular systems studied in this work.

Once that the preferential solvation parameters are determined by fitting viscosity data, the local composition (x_1^L, x_2^L, x_{ij}^L) can be calculated from eqs 9–11:

DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991–15002

$$x_1^{\mathrm{L}} = \frac{(1-x_2)^m}{(1-x_2)^m + g_{2/1}(x_2)^m + g_{ij/1}[(1-x_2)x_2]^{m/2}} \quad (9)$$

$$x_2^{\mathrm{L}} = \frac{g_{2/1}(x_2)^m}{(1-x_2)^m + g_{2/1}(x_2)^m + g_{ij/1}[(1-x_2)x_2]^{m/2}} \quad (10)$$

$$x_{ij}^{\mathrm{L}} = \frac{g_{ij/1}[(1-x_2)x_2]^{m/2}}{(1-x_2)^m + g_{2/1}(x_2)^m + g_{ij/1}[(1-x_2)x_2]^{m/2}} \quad (11)$$

where *ij* subscripts refer to mutual (1−2) interactions or (2−2) self-association interactions.

In summary, the PS viscosity model (eq 8) can be used to correlate experimental data and its fitted parameters ($g_{2/1}$, $g_{ij/1}$, η_{ij}) can be used to estimate trends in the viscosity (e.g., maxima or minima) and to estimate local composition (eqs 9−11) that influence chemical phenomena. Local compositions obtained from the PS viscosity model have been demonstrated to show correspondence with local compositions obtained from molecular simulations and spectroscopic methods.[21]

2.2. Grunberg−Nissan Viscosity Model. The Grunberg−Nissan viscosity model[19,20] is given by eq 12:

$$\ln(\eta_{\mathrm{mix}}) = x_1\ln(\eta_1) + x_2\ln(\eta_2) + x_1x_2G_{ij} \quad (12)$$

where η_{mix}, η_1, and η_2 are the dynamic viscosities of the mixture, pure 1, and pure 2 components, respectively. The G_{ij} is a single adjustable binary interaction parameter related to the activation energy of molecular translation.

2.3. McAllister Viscosity Model. The four-body McAllister viscosity model[14] was developed by considering the activation energy of molecular motion that can be expressed as the interaction of *ij* molecules given by eq 13

$$\begin{aligned}\ln(K_{\mathrm{mix}}) = {} & x_1^4\ln(K_1) + 4x_1^3x_2\ln(K_{1112}) + 6x_1^2x_2^2\ln(K_{1122}) \\ & + 4x_1x_2^3\ln(K_{2221}) + x_2^4\ln(K_2) \\ & - \ln[x_1 + x_2M_2/M_1] \\ & + 4x_1^3x_2\ln[(3 + M_2/M_1)/4] \\ & + 6x_1^2x_2^2\ln[(1 + M_2/M_1)/2] \\ & + 4x_1x_2^3\ln[(1 + 3M_2/M_1)/4] + x_2^4\ln[M_2/M_1]\end{aligned} \quad (13)$$

where K_{mix} and the K_i parameters are the kinematic viscosity of the mixture and pure component *i*, respectively, and *M* is the molar mass. Equation 13 has three fitting parameters, K_{1112}, K_{1122}, and K_{2221} that are related to specific interactions in the molecular system. When eq 13 is applied to molecular solvent−DES mixtures, components 1 and 2 are defined to be the molecular solvent and the DES, respectively.

2.4. Jouyban−Acree Viscosity Model. The Jouyban−Acree model[17] has three fitting parameters (A_0, A_1, A_2) and the following form:

$$\ln(\eta_{\mathrm{mix}}) = x_1\ln(\eta_1) + x_2\ln(\eta_2) + A_0\left[\frac{x_1x_2}{T}\right] + A_1\left[\frac{x_1x_2(x_1-x_2)}{T}\right] + A_2\left[\frac{x_1x_2(x_1-x_2)^2}{T}\right] \quad (14)$$

2.5. Objective Function. To obtain the adjustable parameters in the viscosity models, the average relative deviation (ARD) was used as the objective function:

$$\mathrm{ARD}[\%] = (1/N)\sum |(\eta_{\mathrm{Cal}} - \eta_{\mathrm{Exp}})/\eta_{\mathrm{Exp}}| \times 100 \quad (15)$$

3. RESULTS AND DISCUSSION

3.1. Comparison of Viscosity Correlation Models. Table 1 provides a summary of correlation results for all viscosity models and for all molecular systems studied. In the discussion, the ideal liquid viscosity is defined as η^{ideal}, which is the mole fraction average of the pure component viscosities at constant temperature and pressure. Table S1 (Supporting Information) provides a detailed list of the molecular systems and data sources[3,22,24−100] and contains systems that exhibit viscosity maxima (+S), viscosity minima (−S) or positive (+), negative (−), or positive and negative (+−) deviations in mixture viscosity compared with η^{ideal}. Parameter values shown in Table S1 for the PS viscosity model are discussed later. In Table 1, it can be seen that the PS, McAllister, and Jouyban−Acree models have relatively low (<2%) overall ARD, whereas the Grunberg−Nissan model has high (>7%) overall ARD. In comparing the PS viscosity model with other models, the PS viscosity model has lower ARD for aqueous systems and comparable ARD for the other molecular systems. All models provide low ARD (<1%) for nonpolar molecule−polar molecule systems.

Figure 1 shows typical results for correlation of selected binary mixtures (Figure 1a−e) for all viscosity models. The PS viscosity model conforms remarkably well to aqueous system data (Figure 1a) while the Grunberg−Nissan model provides a qualitative representation of systems that have a viscosity maximum (Figure 1a, b). All models provided reliable correlation of nonpolar−polar (Figure 1c), molecular solvent−IL (Figure 1d) and molecular solvent−DES (Figure 1e) mixtures, with the Jouyban−Acree model being able to conform most closely to the data than the other models.

Plots of the ARD in viscosity for all models versus composition for all systems are given in the Supporting Information (Figures S1−S4) along with parity plots (Figures S5-S8). In general, the PS viscosity model had the lowest ARD among the models examined for aqueous systems but higher ARD than the McAllister or Jouyban−Acree models for other molecular systems. The reason for the higher ARD of the PS viscosity model than the other models for molecular solvent−IL and molecular solvent−DES systems might be due to neglect of mutual (1−2) interactions (eq 7). The (1−2) interactions can be included in eq 5 to improve the correlation; however, this increases the number of parameters and reduces the reliability of their values according to the number of data. The PS model was compared with the other viscosity models using a minimal number of parameters, namely, eq 7 was used for molecular solvent−IL and molecular solvent−DES systems.

3.2. Preferential Solvation Parameters. The values of the fitted parameters, $g_{2/1}$, $g_{12/1}$ and either η_{12} or η_{22}, in the PS viscosity mode at 25 °C are shown in Table S1 (Supporting Information). PS viscosity model parameter values determined

DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991−15002

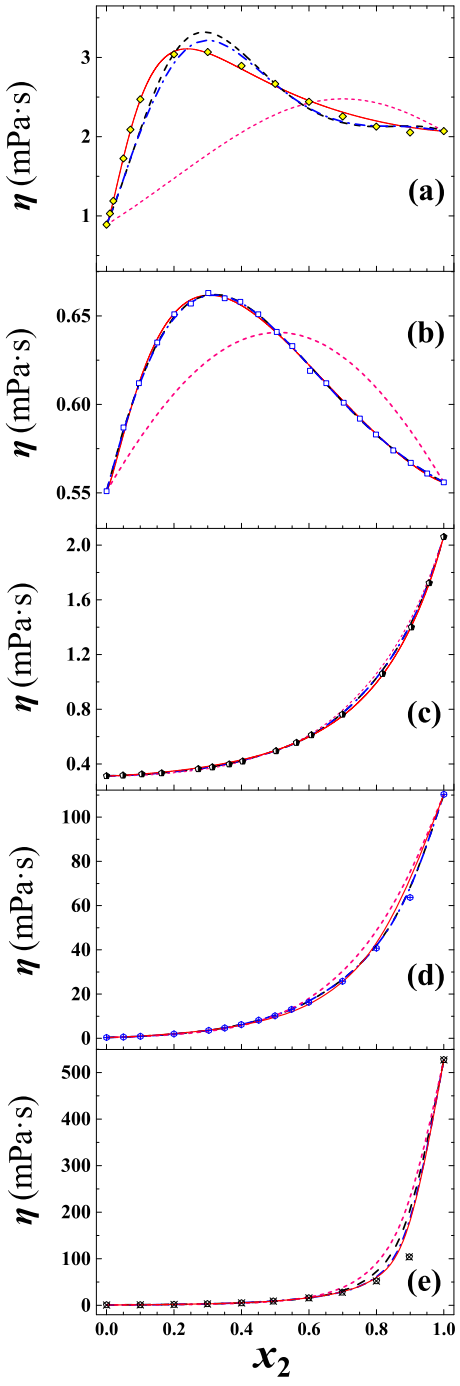


Figure 1. Correlation of dynamic viscosities (η) for selected 1−2 binary mixtures. Symbols: (a) water−2-propanol, (b) methanol−chloroform, (c) hexane−2-propanol, (d) methyl acetate−[bmim][BF_4], and (e) water−choline chloride:urea in 1:2 ratio. Red solid line: preferential solvation viscosity model (eq 8). Blue dotted−dashed line: McAllister model (eq 13). Black long dashed line: Jouyban−Acree model (eq 14). Pink short dashed line: Grunberg−Nissan model (eq 12). Viscosity data references are given in Table S1 (Supporting Information).

at higher temperatures are given in Table S2. Interaction parameters for other models are given in Table S3. Analysis of the parameters of the PS model (Table S1) can provide some insight into the molecular interactions and the solution structure in the molecular systems.[21]

For aqueous molecular systems (Table S1, nos. 1−14), when $g_{2/1}$ values are larger than unity, an S_2 molecule preferably occupies the center of an arbitrary cybotactic region and is surrounded by S_1 (H_2O) molecules. Similarly, when $g_{12/1}$ values are larger than unity, an S_{12} (complex) molecule occupies the center of an arbitrary cybotactic region and is surrounded by S_1 (H_2O) molecules. For a given binary mixture among the aqueous molecular systems, the value of η_{12} relative to the pure component η_1 or η_2 values indicate positive synergistic molecular interactions, because η_{12} is greater than either the η_1 or η_2 values. All aqueous molecular systems shown have positive synergistic molecular interactions leading to viscosity maxima that are related to hydration shell formation. For aqueous mixtures of an HBA solvent, the size of the hydration shell around the S_{12} complex molecules is proportional to the magnitude of the η_{12} values.[21] The presence of hydrophobic functional groups in a solute molecule increases the size of the hydration shell in aqueous mixtures,[101] and correspondingly the values of η_{12} and $g_{2/1}$ for aqueous mixtures of methanol, ethanol, 1-propanol or 2-propanol in water (Table S1, nos. 11−14) tend to increase with the alcohol alkyl chain length.

For nonaqueous molecular systems (Table S1, nos. 15−53), some binary liquid mixtures have positive or negative synergistic molecular interactions, but most systems show negative deviations from η^{ideal}. The strength of the interactions in nonaqueous molecular systems are generally weaker than those in aqueous molecular systems as evident by the relative size of the $g_{2/1}$, $g_{12/1}$, and η_{12} parameters compared with aqueous systems (Table S1), since a solvation shell, if it exists, contains loosely bound molecules.[21,31] The $g_{2/1}$ values in nonaqueous molecular systems (Table S1) are generally less than unity meaning that an S_1 molecule (methanol or ethanol) preferably occupies the center of an arbitrary cybotactic region and is surrounded by S_2 molecules. The η_{12} parameter values are generally less than or intermediate to the pure component viscosities, which indicates negative deviations from η^{ideal}. Negative synergistic molecular interactions can be for methanol−dimethyl carbonate (Table S1, no. 23) and a number of ethanol−HBA molecular systems (Table S1, nos. 34−37, 41, 43, 48) in which complex molecule specific interactions are strong (high $g_{12/1}$ values) and η_{12} values are less than either η_1 or η_2.

For nonpolar−polar molecular systems (Table S1, nos. 54−77), self-association is the chief specific interaction (ij = 22) and all viscosity values had negative deviations from η^{ideal}. In general, the $g_{2/1}$ values were less than unity and the η_{22} values were intermediate to η_1 or η_2, although there are a few exceptions (Table S1, nos. 54, 71, 73) due to the presence of a minimum in viscosity.

For molecular solvent−IL and molecular solvent−DES systems (Table S1, nos. 78-103), the solutions are made up of a low viscosity compound (1) and a high viscosity compound (IL or DES) such that the $g_{2/1}$ values tend to be small indicating that the molecular solvent tends to be surrounded by the IL or DES. Based on the PS model parameters, the molecular solvent tends to occupy hydrophilic or hydrophobic free volume regions in the IL or DES framework. In general (Table S1), the η_{22} values are intermediate to η_1 or η_2 and the $g_{22/1}$ values are larger than the $g_{2/1}$ values implying that there is strong self-interaction of the IL or DES, thus justifying the use of eq 7.

3.3. Relationship of PS Parameters with Other Viscosity Model Parameters. It is of interest to examine the correlation of parameters of the PS viscosity model with parameters of other models. No clear correlation could be found between the parameters of the PS model and other models for

DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991−15002

molecular solvent−IL or molecular solvent−DES systems. On the other hand, the PS viscosity model parameters and McAllister model parameters were found to correlate for aqueous, nonaqueous, and nonpolar−polar molecular systems as shown in Figure 2. In aqueous systems, the η_{12} parameter

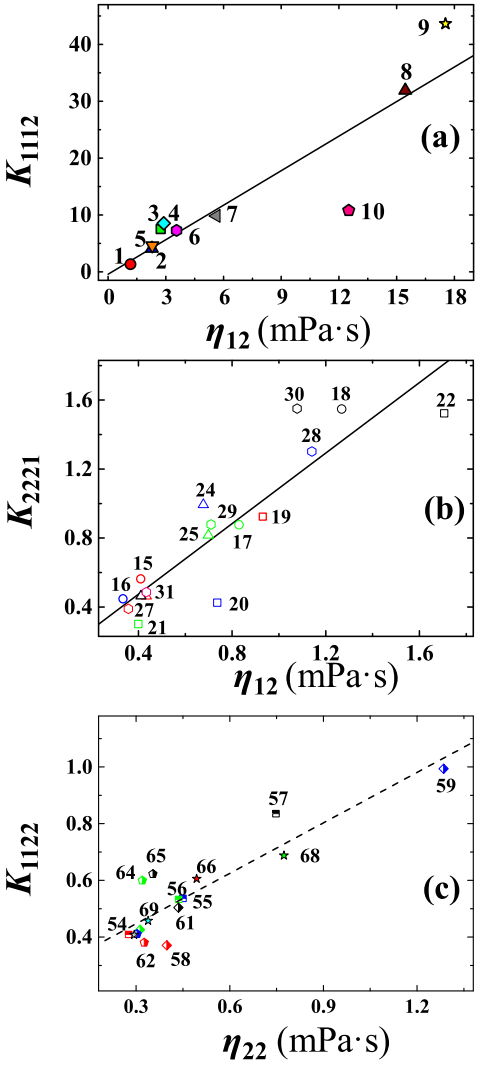


Figure 2. Relationships between preferential solvation parameters (η_{12} or η_{22}, eq 8) and McAllister binary interaction parameters (K_{1112}, K_{1122}, K_{2221}, eq 13) for (a) aqueous systems (R^2 = 0.83 and N = 10), (b) nonaqueous systems (R^2 = 0.83 and N = 17), and (c) nonpolar−polar systems (R^2 = 0.81 and N = 17) at 25 °C. Symbols and numbers are listed in Table S1. Aqueous systems in part a are H_2O− dipolar protophilic mixtures (nos. 1−10, Table S1). Nonaqueous systems in part b are mixtures of methanol (nos. 15−31, Table S1). Nonpolar−polar systems in part c are mixtures of hexane (nos. 54−69, Table S1).

showed correlation with the K_{1112} parameter (Figure 2a), which is attributed to the presence of a hydration shell formation that is related to the K_{1112} parameter. On the other hand, due to lack of a solvation shell in nonaqueous systems (Figure 2b), the η_{12} parameter showed correlation with the K_{2221} parameter. The fitting parameters of other models (Jouyban−Acree and Grunberg−Nissan) showed low correlation with the η_{12} parameter of the PS viscosity model (R^2 < 0.80, Figure S9, Supporting Information) that is attributed to the empirical nature of those models.

The four-body interaction form of the McAllister model[14] is required to describe systems with maxima in viscosity such as water−acetone or maxima and minima in viscosity such as methanol and toluene.[102] The PS viscosity model can describe either maximum viscosity or minimum viscosity behavior well, and it can also describe both positive and negative deviations from η^{ideal} as evident by results for methanol−anisole, methanol−dichloromethane, and ethanol−chloroform in Table S1 (nos. 17, 21, 38). The PS viscosity model cannot describe molecular systems that have both a maximum and a minimum in mixture viscosity outside of the pure component viscosity values using only (1−2) or (2−2) interactions. However, by including additional terms (e.g., (1−2), (2−2), or both (1−2) and (2−1)), it may be possible to represent mixture viscosity of systems that have both a maximum and a minimum.

4. APPLICATIONS

PS viscosity model parameters in Table S1 can be used with eqs 9−11 to estimate the local composition in chemical systems. The local composition provides the population of complex

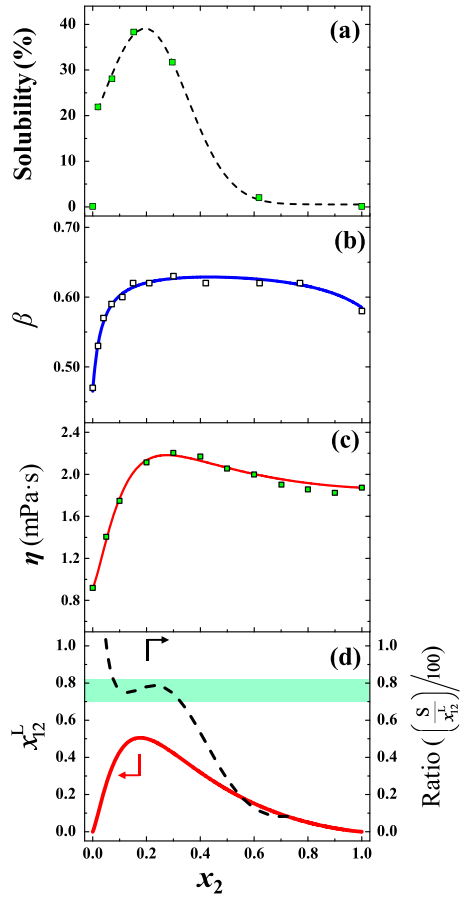


Figure 3. Plots of (a) solubility of lignin, (b) Kamlet−Taft basicity (β), (c) dynamic viscosity (η), and (d) local composition of (1−2) complexes (x_{12}^L) and ratio of the solubility and local (1−2) complex composition (S/x_{12}^L) versus γ-valerolactone (GVL) mole fraction (x_2) in the water (1) and GVL (2) mixtures at 25 °C, except for solubility data obtained at 40 °C. Solubility data was taken from the literature.[2] The dashed line in part a was calculated from a Gaussian equation. The β values in part b were obtained from the literature.[116] The green-highlighted area in part d shows direct proportion of S/x_{12}^L values at maximum solubility.

DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991−15002

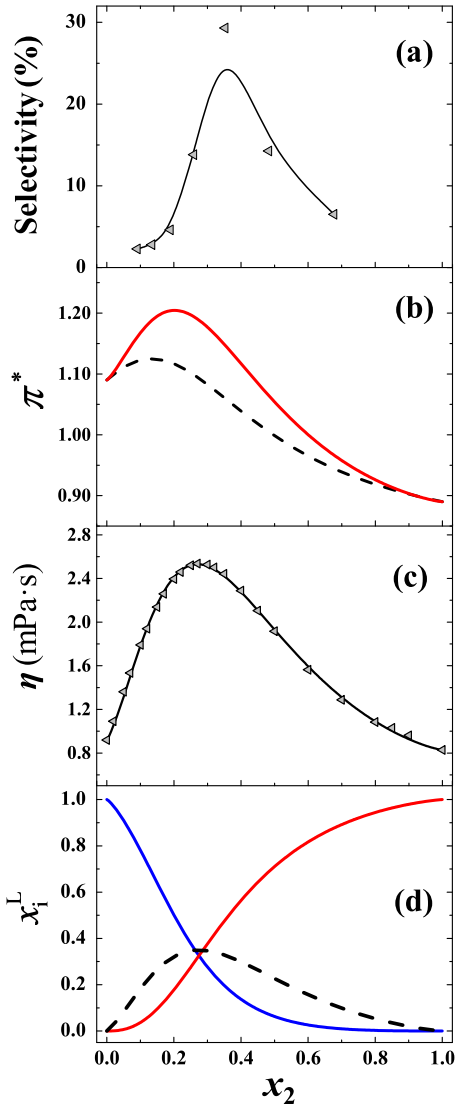


Figure 4. Plots of (a) selectivity of carbazole from crude anthracene, (b) dipolarity/polarizability (π^*), (c) dynamic viscosity (η), and (d) local compositions (x_i^L, eqs 9−11) versus dimethylformamide (DMF) mole fraction (x_2) in the water (1) and DMF (2) mixtures at 25 °C, except for selectivity that obtained at 30 °C. The line in part a aids the eye. Solid lines in part b are calculated from the PS viscosity model with eq 16. The π_1^*, π_2^*, and π_{12}^* values used in eq 16 were (1.09, 0.89, 1.56). The dashed line in part b is spectra data.[117] Lines in part d: x_1^L (blue line), x_2^L (red line), and x_{12}^L (dashed line). Selectivity data was taken from the literature.[107]

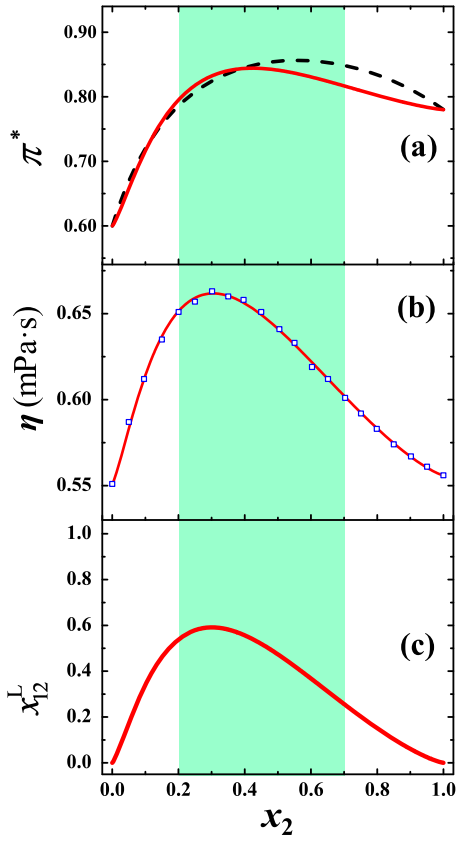


Figure 5. Plots of (a) dipolarity/polarizability (π^*), (b) dynamic viscosity (η), and (c) local composition of (1−2) complexes (x_{12}^L) versus chloroform mole fraction (x_2) in the methanol (1) and chloroform (2) mixtures at 25 °C. Solid lines calculated from the PS viscosity model with eqs 8 and 16. The π_1^*, π_2^*, and π_{12}^* values used in eq 16 were (0.60, 0.73, 0.93). The dashed line is spectra data.[118] The green-highlighted area shows bulk compositions for applications.

molecules or associated or free mobile molecules that are related to chemical phenomena in reaction and separation processes. Thus, the range of bulk compositions appropriate for a given process can be used to estimate the variation of solvation effects in the underlying chemical phenomena. Four applications are considered, and emphasis is placed on aqueous systems, since the PS viscosity model provides local compositions that correspond well with theoretically and spectroscopically determined values for aqueous systems.[21]

4.1. Lignin Fractionation with Aqueous Molecular Systems. Water−γ-valerolactone (GVL) mixtures have been proposed for fractionating lignin.[103,104] The solubility of lignin changes with the composition of the solvent mixture (Figure 3a), and it increases greatly when GVL is present (Figure 3b), because lignin dissolution depends on the Kamlet−Taft basicity (β) of the solvent mixture.[105,106] While β does show significant variation with composition in water-rich regions, it cannot explain the decrease in lignin solubility with an increasing GVL composition at $x_2 > 0.2$. On the other hand, parameters of the PS viscosity model can be determined from available viscosity data (Figure 3c) and then used to estimate the local composition of the (1−2) complex molecules (Figure 3d), where it is seen that the population of (1−2) complex molecules is directly proportional to the lignin solubility (Figure 3d). For example (green highlight, Figure 3d), the ratio of the solubility and local (1−2) complex composition (S/x_{12}^L) was approximately 0.8 at maxima solubility (Figure 3a, $x_2 \approx 0.1$−0.3).

4.2. Carbazole Separation with Aqueous Molecular Systems. Separation of carbazole from crude anthracene typically requires a solvent mixture to vary solvent selectivity (Figure 4a)[107] via solvent polarity.[108−110] Solvent polarity can be measured with spectroscopic methods[25,31] (Figure 4b); however, π_V^* can be estimated as in Figure 4b from viscosity measurements (Figure 4c). The PS viscosity model gives local compositions of (1), (2), and (1−2) species (Figure 4d), so by using one spectral measurement of the solvent mixture (π_{12}^*) and available spectral values of the pure solvents (π_1^*, π_2^*), we find π_V^* via

$$\pi_V^* = x_1^L\pi_1^* + x_2^L\pi_1^* + x_{12}^L\pi_{12}^* \quad (16)$$

DOI: 10.1021/acs.iecr.9b01179
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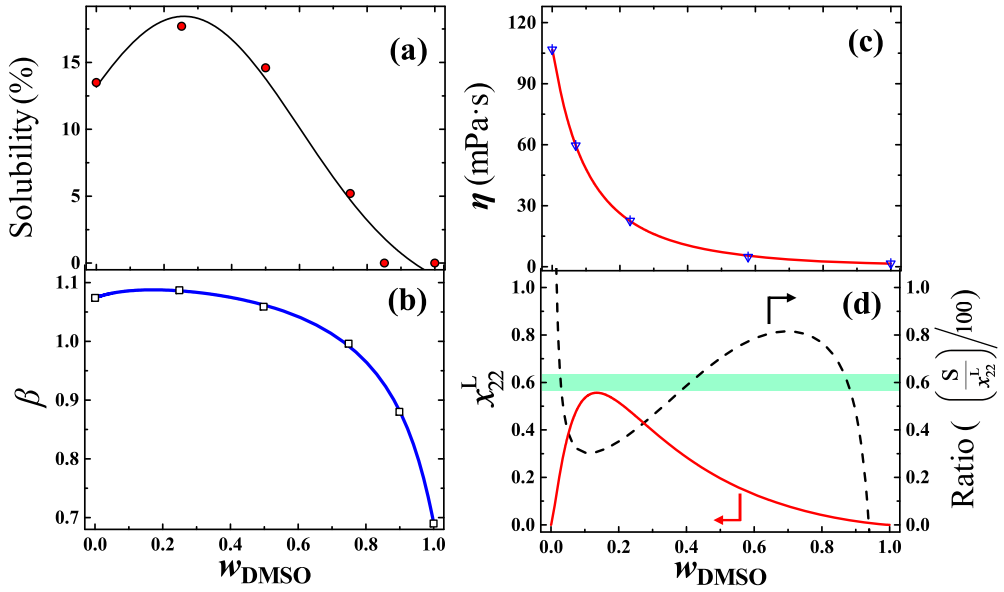


Figure 6. Mixtures of dimethyl sulfoxide (DMSO) and1-ethyl-3-methylimidazolium diethyl phosphate ([EMIm][DEP]) showing (a) solubility of cellulose in term of mass fractions at 60 °C and (b) Kamlet−Taft basicity (β) at 25 °C as a function of weigh fraction (w_{DMSO}). Mixtures of DMSO and 1,3-dimethyl imidazolium dimethyl phosphate ([MMIm][DEP]) showing (c) dynamic viscosity (η) at 40 °C and (d) local composition of self-association complexes (x_{22}^{L}) ratio of the solubility and local (2−2) complex composition (S/x_{22}^{L}). Solubility data and β values in parts a and b were obtained from the literature.[114] The dashed line in part a was calculated from the Gaussian equation. The viscosity data at 40 °C in part c was taken from the literature.[92] The green-highlighted area in part d shows an average of the S/x_{22}^{L} values.

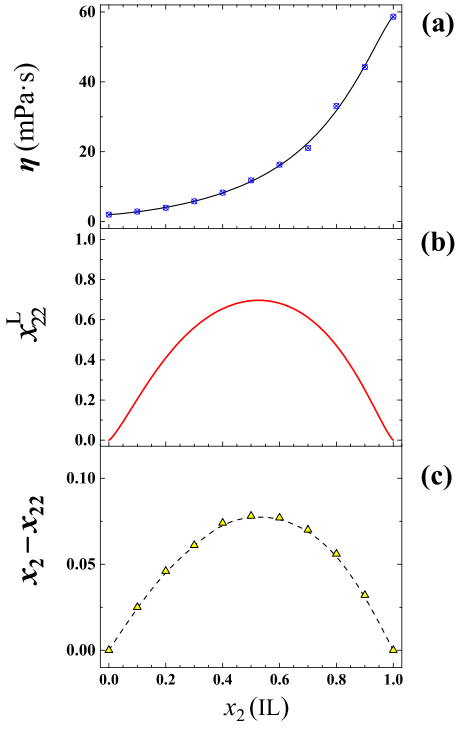


Figure 7. Mixtures of dimethyl sulfoxide (1)−diethylamine acetate (2) showing: (a) dynamic viscosity (η), (b) viscosity-derived local composition of self-association complexes (x_{22}^{L}), and (c) extent of local composition of diethylamine acetate (x_2-x_{22}) at 25 °C. Solid lines in parts a and b were calculated with the PS viscosity model using eqs 8 and 11, respectively. The extent of local composition data were taken from the literature.[22]

from viscosity-derived local composition (Figure 4d). The advantage of eq 16 over spectral measurements is that it provides appropriate values of the local composition without the influence of dyes or indicators that show bias toward one of the solvents in the solvent mixture. Alternatively, the population of the (1−2) complex molecules (Figure 4d) may be used directly to imply changes in solvent mixture selectivity to avoid having to make the spectral measurements altogether. Namely, the selectivity of the solvent mixture should be examined at several compositions (*e.g.* x_2 = 0.1, 0.3, and 0.6) according to the trend of the local composition of the (1−2) complex (Figure 4d).

4.3. Chromatographic Analysis with Nonaqueous Molecular Systems. Mixtures of methanol and chloroform are typically used in chromatographic analyses[111,112] and for small-scale lipid extractions.[113] In some applications, a chloroform mole fraction of x_2 = 0.2 is used[113] with the reason being related to solvent polarity (Figure 5a). Through the use of eq 16, it is possible to estimate the solvent polarity over the full composition range using a minimal amount of spectral data and by using local compositions determined from viscosity data (Figure 5b). Alternatively, one may estimate possible bulk composition regions by considering the population of (1−2) complexes given by the PS viscosity model (Figure 5c) as shown by the green-shaded region in Figure 5.

4.4. Cellulose Dissolution with Molecular Solvent−IL Systems. Dimethyl sulfoxide (DMSO) can be added to 1-ethyl-3-methylimidazolium diethyl phosphate ([EMIm][DEP]) ionic liquids to lower solution viscosity and to increase cellulose solubility (Figure 6a).[114] The phenomena is explained in the literature[114] by the increase in Kamlet−Taft basicity (β) of the solvent mixture (Figure 6b). However, the Kamlet−Taft β values do not change significantly when DMSO is added to the IL up to weight fractions of DMSO as high as 0.5 (Figure 6b). On the other hand, by fitting the PS viscosity model to DMSO-1,3-dimethyl imidazolium dimethyl phosphate ([MMIm][DEP] viscosity data (Figure 6c), which is similar in chemical nature to [EMIm][DEP], local composition of associated IL species (x_{22}^{L}) can be determined (Figure 6d) that provides the

DOI: 10.1021/acs.iecr.9b01179
Ind. Eng. Chem. Res. 2019, 58, 14991−15002

population of mobile (2−2) species surrounded by DMSO molecules. The population of (2−2) species (Figure 6d) have a direct effect on the cellulose solubility and are most likely responsible for the experimentally observed maximum solubility when DMSO is added to [MMIm][DEP] ionic liquid. Namely, the maximum population of mobile associated (2−2) species in DMSO−IL mixtures allows the associated (2−2) species to interact with cellulose. The solubility to local composition ratio (S/x_{22}^{L}) was approximately 0.5 to 0.6 over most of the composition region (Figure 6d, green highlight) showing that the solubility is proportional to the population of associated (2−2) species. Considering the solubility of cellulose in pure DMSO, addition of IL (Figure 6) causes an increase in Kamlet−Taft β values such that cellulose dissolution phenomena can be explained by the trend in β. The minimum β required might be used with the corresponding x_{22}^{L} values from the PS viscosity model to estimate ranges of bulk composition that are appropriate for cellulose dissolution in a given organic solvent.

Viscosities of DMSO (1)−diethylamine acetate (2) mixtures are shown in Figure 7a, in which the x_{22}^{L} values (Figure 7b) obtained from the PS viscosity model were compared with the extent of local compositions (x_2-x_{22}, Figure 7c) reported in the literature.[22] Trends in both x_{22}^{L} and x_2-x_{22} showed slight skewness in IL-rich compositions at $x_2 \approx 0.60$ and their values were in close correspondence. Although (1−2), (2−2), and multibody interactions exist in molecular solvent−IL systems, the assumption of using only (2−2) self-interactions in the PS viscosity model to estimate local composition appears to be valid (Figure 7). Molecular simulations[23] of molecular solvent−IL systems also show that the (2−2) self-interactions are much stronger than the (1−1) or (1−2) interactions in support of using only (2−2) self-interactions in eq 8. Agreement between PS viscosity-derived local compositions and spectral-derived local compositions (Figure 7) infers that the assumption of using (2−2) associated complexes in eq 8 can provide qualitative estimation of local composition trends in molecular solvent−IL mixture systems. In addition to molecular solvent−IL systems, the PS viscosity model is applicable to polymer solution−IL systems, in which viscosity-derived local compositions correlate with spectral-derived microviscosities[115] for the polyethylene glycol (PEG)−[Bmim][PF_6] system (Supporting Information, Figure S10).

5. CONCLUSIONS

In this work, the preferential solvation (PS) viscosity model was used to correlate viscosities of five types of molecular systems, where it was found that the PS viscosity model gave lower deviations for aqueous systems than several commonly used viscosity models. The PS viscosity model was found to correlate liquid mixture viscosity data (2549 points) for five types of chemical systems to within an average relative deviation of 1.7%, compared with Grunberg−Nissan (7.6%), McAllister (1.6%), and Jouyban−Acree (1.7%) models. The PS viscosity model cannot describe chemical systems that have both a maximum and a minimum in viscosity with composition (methanol−toluene) using only (1−2) interactions so that it is likely that additional interactions are needed to describe such systems. The $g_{2/1}$ and η_{ij} parameters obtained from PS model are useful in analyzing the variation of viscosity with composition. Qualitative comparison of PS viscosity model parameters with pure component property viscosities allows one to readily recognize synergism in viscosity with composition, negative or positive deviations from ideal liquid viscosity, relative strength of molecular interactions and relative size of hydration or solvation structures. Solubilities of lignin in solvent mixtures and solubilities of cellulose in solvent mixtures were found to be directly proportional to the population of (1−2) or (2−2) complexes, respectively, as determined from the PS viscosity model. The PS viscosity model is useful for explaining solvent mixture composition trends in example applications related to lignin solubility, carbazole separation, chromatographic analyses, and cellulose dissolution. The PS viscosity model is widely applicable to chemical systems and can be used to relate local composition effects to chemical phenomena.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.9b01179.

Tables S1−S3 tabulate binary interaction parameters for four viscosity model. Figures S1−S4 and S5−S8 show ARD values and parity plots, respectively, for all viscosity models. Figure S9 provides relationships between preferential solvation parameters and Jouyban−Acree binary interaction parameters. Figure S10 shows spectral-derived microviscosities for the PEG−[Bmim][PF_6] system (PDF)

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The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Partial support of this research by JSPS KAKENHI (R.L.S.), Grant-in-Aid Scientific Research (B) Number 16H04549 is gratefully acknowledged.

■ ABBREVIATIONS

Molecular Solvent

Ace = acetone
ACN = acetonitrile
Ans = anisole
BuAc = *n*-butyl acetate
BuOH = 1-butanol
CHN = cyclohexanone
$CHCl_3$ = chloroform
CIN = 1,8-cineole
CPN = cyclopentanone
CS_2 = carbon disulfide
DCM = dichloromethane
DMA = *N,N*-dimethylacetamide
DMC = dimethyl carbonate
DEC = diethyl carbonate
DME = dimethoxyethane
DMF = *N,N*-dimethylformamide

DOI: 10.1021/acs.iecr.9b01179
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DMSO = dimethyl sulfoxide
EtOH = ethanol
EtAc = ethyl acetate
FA = formamide
GBL = γ-butyrolactone
GVL = γ-valerolactone
HeOH = 1-hexanol
Hep = heptane
Hex = hexane
iAAc = isoamyl acetate
MeAc = methyl acetate
MeOH = methanol
MEK = 2-butanone
MIBK = methyl isobutyl ketone
NMP = *N*-methyl-2-pyrrolidone
PEG200 = polyethylene glycols with averaged molecular weight of 200
PEG400 = polyethylene glycols with averaged molecular weight of 400
PEG600 = polyethylene glycols with averaged molecular weight of 600
1-PrOH = 1-propanol
2-PrOH = 2-propanol
PPC = propylene carbonate
Pyr = pyridine
THF = tetrahydrofuran

Cation

[amim] = 1-allyl-3-methylimidazolium
[bmim] = 1-butyl-3-methylimidazolium
DEAA = diethylamine acetate
[emim] = 1-ethyl-3-methylimidazolium
[mmim] = 1,3-dimethyl imidazolium

Anion

AC = acetate
[BF_4] = tetrafluoroborate
Cl = chloride
[Ntf_2] = bis(trifluoromethylsulfonyl)imide
[PF_6] = hexafluorophosphate

Deep Eutectic Solvent

[Ch]Cl = choline chloride
ethaline = choline chloride−ethylene glycol (1:2 molar ratio)
ETG = ethylene glycol
glyceline = choline chloride−glycerol (1:2 molar ratio)
HBA = hydrogen bond acceptor
HBD = hydrogen bond donor
NaCl = sodium chloride
NABr = sodium bromide
NaI = sodium iodide
LA = lactic acid
reline = choline chloride−urea (1:2 molar ratio)
TEG = tetraethylene glycol

Latin Symbols

A_0, A_1, A_2 = Jouyban−Acree binary interaction parameters, eq 14,
$g_{2/1}$ = viscosity-based preferential solvation parameter, eq 8
$g_{ij/1}$ = viscosity-based solvation parameter, eq 8
G_{ij} = Grunberg−Nissan binary interaction parameter, eq 12
K = kinematic viscosity
K_{1112}, K_{1122}, K_{2221} = McAllister binary interaction parameters, eq 13
M = molar mass
m = number of molecules in the local region, according to eq 8
S/x_{12}^{L} = ratio of the solubility and local (1−2) complex composition
x = bulk mole fraction
x_2-x_{22} = extent of local compositions

Greek Symbols

β = Kamlet−Taft basicity
η = dynamic viscosity
η_{12} = viscosity of the mutual (1−2) complex molecule, eq 8
η_{22} = viscosity of the self (2−2) complex molecule, eq 8
η^{ideal} = mole fraction average of the pure component viscosities
π^* = spectroscopic Kamlet−Taft dipolarity/polarizability
π_V^* = viscosity-derived Kamlet−Taft dipolarity/polarizability

Superscript

L = local composition

Subscript

1 = component 1
2 = component 2
12 = mutual complex solvent molecule pair
22 = self-complex solvent molecule pair
max = maximum
mix = mixture

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